COVID-19 Chest CT Image Segmentation

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I. Introduction

Coronavirus disease (COVID-19) created a major global health challenge and increased the demand for fast and reliable medical diagnosis. Chest X-ray (CXR) imaging is widely available, low-cost, and commonly used to examine lung conditions. In many hospitals, X-ray images are used as a supportive tool to assess respiratory infections, including COVID-19. However, manual inspection of X-rays requires expert radiologists and can be time-consuming, especially when the number of patients is large.

Deep learning has shown strong performance in medical image analysis, especially for segmentation tasks. Image segmentation aims to classify each pixel of an image, which makes it possible to identify and localize abnormal regions. In the context of COVID-19, infection segmentation helps highlight the infected area in the lungs, which can assist clinicians in disease monitoring, severity estimation, and treatment planning.

In this labwork, we focus on segmenting infection regions in COVID-19 chest X-ray images using the COVID-QU-Ex dataset. First, we download and explore the dataset to understand its structure and the available images and masks. Then, we build and train one deep learning segmentation model (U-Net) to predict infection masks from input X-ray images. Finally, we evaluate the model performance using segmentation metrics and visualize predicted masks to verify the segmentation quality.

II. Dataset

In this labwork, we use the COVID-QU-Ex dataset. It contains chest X-ray (CXR) images collected for three categories:COVID-19, Non-COVID, and Normal. The dataset also provides segmentation masks so that we can perform infection segmentation.

The dataset is organized into three main splits:

- **Train**: used to train the model
- **Validation (Val)**: used to tune the model and monitor performance during training
- **Test**: used for final evaluation

For each split and each class, the dataset includes:

- an **images** folder: the input chest X-ray images
- an **infection masks** folder: the ground-truth binary masks, where white pixels represent infection regions and black pixels represent background

It is important to note that many samples in the Normal (and some Non-COVID class have masks that are completely black, meaning no infection is present. This is expected and reflects real clinical conditions.

Before training, the images and masks are resized to a fixed size (e.g., **256**×**256**) and normalized. These steps help the model learn efficiently and ensure all inputs have the same shape.

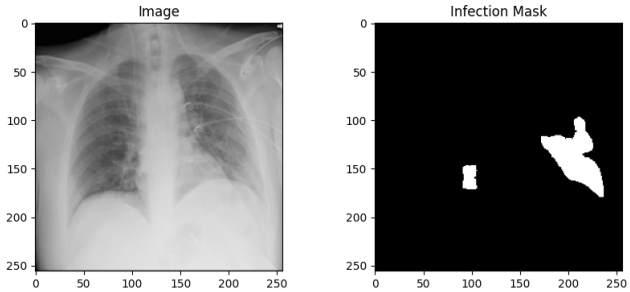

Fig. 1. Sample dataset

Figure 1 shows a sample chest CT image along with its corresponding infection mask. The infection mask is a binary image where white pixels indicate infected regions and black pixels represent non-infected areas. As observed, infection regions typically occupy a relatively small portion of the image, highlighting the class imbalance problem at the pixel level. This imbalance makes segmentation more challenging and justifies the use of Dice-based loss functions during training.

III. Methodology

A. Problem Formulation

The task is formulated as a *binary semantic segmentation* problem. Given an input chest CT image $X \in R^{H \times W \times 3}$, the goal is to predict a binary infection mask $Y \in \{0,1\}^{H \times W}$, where 1 denotes infected pixels and 0 denotes background pixels.

B. Data Preprocessing

All images and corresponding infection masks were preprocessed as follows:

- Images and masks were resized to 256×256 pixels to standardize the input size.
- Image intensities were normalized to the range $[0, 1]$ by dividing by 255.
- Infection masks were converted to binary values (0 for background and 1 for infection).

The official dataset split (Train/Val/Test) was used. Lung masks provided by the dataset were not utilized in this work.

C. Segmentation Model

A U-Net architecture with a ResNet34 encoder pretrained on ImageNet was adopted for infection segmentation. The encoder extracts multi-scale feature representations while progressively downsampling spatial resolution. The decoder performs upsampling to recover spatial details and uses skip connections to fuse high-resolution features from the encoder with the decoder features at corresponding scales, improving boundary localization. The final prediction layer applies a 1×1 convolution to produce a single-channel logit map for pixel-wise classification.

D. Loss Function

Infection regions typically occupy a small portion of an image, leading to strong class imbalance. To address this, we used a combination of Dice loss and Binary Cross-Entropy (BCE) loss:

$$\mathcal{L} = \mathcal{L}_{Dice} + \mathcal{L}_{BCE}. \tag{1}$$

The Dice coefficient is defined as:

$$\text{Dice} = \frac{2TP}{2TP + FP + FN}, \tag{2}$$

where TP, FP, and FN denote true positive, false positive, and false negative pixel counts, respectively. Dice loss is defined as $\mathcal{L}_{Dice} = 1 - \text{Dice}$.

E. Training Strategy

The model was trained using the Adam optimizer with learning rate 1×10^{-4}, batch size 8, and 10 epochs. At each iteration, the network outputs a logit map, which is used to compute the combined loss. Validation performance was monitored after each epoch to assess generalization.

F. Evaluation Metrics

Model performance was evaluated using the Dice score, which measures the overlap between the predicted infection mask and the ground-truth mask:

$$\text{Dice} = \frac{2TP}{2TP + FP + FN}. \tag{3}$$

In addition, training and validation losses were tracked across epochs to analyze convergence and detect potential overfitting.

IV. Results

A. Training Performance

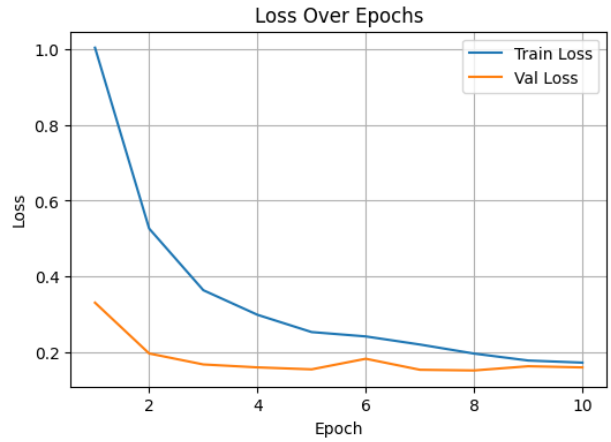

Fig. 2. Training and validation loss over 10 epochs.

Figure 2 shows the training and validation loss curves over 10 epochs. The training loss decreased significantly from approximately 1.00 in the first epoch to around 0.17 in the final epoch, indicating stable convergence of the model. Similarly, the validation loss decreased from approximately 0.33 to 0.16, suggesting improved generalization performance.

The gap between training and validation loss remains relatively small throughout training, indicating that severe overfitting did not occur.

B. Dice Score Evaluation

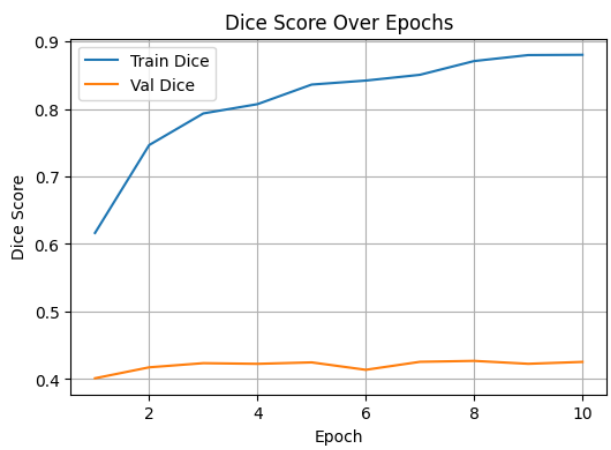

Fig. 3. Training and validation Dice score over 10 epochs.

Figure 3 presents the Dice score progression during training. The training Dice score improved steadily from 0.62 to approximately 0.88 after 10 epochs. This demonstrates that the model successfully learned to segment infection regions from the training data.

The validation Dice score increased slightly from 0.40 to approximately 0.42–0.43 and then stabilized. Although validation performance improved compared to the initial epochs, a noticeable gap between training and validation Dice scores remains. This gap may be attributed to class imbalance and the limited size of infection regions within the dataset.

C. Discussion of Results

The training curves indicate that the model converges properly and learns meaningful segmentation patterns. However, the relatively lower validation Dice score suggests that generalization performance is limited. This may be caused by:

- Severe pixel-level class imbalance (small infection areas),
- Presence of many normal images with no infection,
- Limited dataset size.

Future improvements could include stronger data augmentation, longer training, or using advanced architectures such as Attention U-Net.

V. References

Ahmet Ilhan, Kezban Alpan, Boran Sekeroglu, Rahib Abiyev, COVID-19 Lung CT image segmentation using localization and enhancement methods with U-Net, Procedia Computer Science, Volume 218, 2023, Pages 1660-1667, ISSN 1877-0509, https://doi.org/10.1016/j.procs.2023.01.144.